PROJECT REPORT OF CO-OP Project at Industry (Module-2)

ON

Comparative Analysis of Machine Learning Algorithms for Heart Disease Prediction

submitted in partial fulfilment of the requirements for the award of degree of

BACHELOR OF ENGINEERING

In

COMPUTER SCIENCE AND ENGINEERING

Submitted by:

Tamanna Kansal

2210992436

Supervised By:

Isha Kansal

Assistant Professor

Chitkara University

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

CHITKARA UNIVERSITY INSTITUTE OF ENGINEERING AND TECHNOLOGY
CHITKARA UNIVERSITY, PUNJAB, INDIA

CONTENTS

S.No.	Title	Page No.
1	Abstract	5
2	Introduction	6
3	Methodology	7
4	Tools and Technologies	9
5	Implementation	10
6	Major Findings/Outcomes/Output/Results	11
7	Conclusion and Future Scope	14
8	References	15

DECLARATION

I hereby certify that the work which is being presented in the project report entitled "Comparative Analysis of Machine Learning Algorithms for Heart Disease Prediction" in partial fulfilment of requirement for the award of the degree of Bachelor of Engineering (Computer Science and Engineering) submitted in the department of Computer Science and Engineering at Chitkara University Institute of Engineering and Technology, Chitkara University, Punjab, India, is an authentic record of my own work carried out under the supervision of Isha Kansal. The matter presented in this project report has not been submitted in any other university/institute for the award of any degree.

Place: Rajpura **Tamanna Kansal**

Date: 04-05-2026 2210992436

This is to certify that the above statement made by the candidate is correct to the best of my knowledge and belief.

Isha Kansal

Assistant Professor

Department of Computer Science and Engineering

Chitkara University Institute of Engineering and Technology,

Chitkara University, Punjab, India

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to my supervisor, Ms. Isha Kansal, Assistant Professor, Department of Computer Science and Engineering, Chitkara University, for her invaluable guidance, constant support, and constructive feedback throughout the duration of this project. Her expertise and encouragement greatly contributed to the successful completion of this work.

I am also grateful to the faculty and staff of the Department of Computer Science and Engineering, Chitkara University Institute of Engineering and Technology, for providing the necessary resources and a conducive research environment.

I extend my sincere thanks to the UCI Machine Learning Repository for making the heart disease dataset publicly available, which served as the foundation of this research.

Finally, I express my deepest gratitude to my family and friends for their unwavering support, patience, and encouragement throughout this academic journey.

Tamanna Kansal
2210992436

ABSTRACT

Cardiac disease's early identification plays a vital role in reduced mortality and improvements in patient care. The heart disease diagnostic process requires multiple tests and physician assessments over the patient's record, which is often time-consuming. Artificial intelligence allows for quicker, more precise decisions through prediction assistance with clinical professionals.

This research provides a comparative analysis of five supervised machine learning classifiers — Logistic Regression, Decision Tree, Random Forest, Support Vector Machine (SVM) and XGBoost — for heart disease prediction. The experiments were conducted using the UCI Machine Learning Repository dataset of 1025 samples containing 13 clinical attributes. Preprocessing included missing value imputation, feature scaling using StandardScaler, and an 80/20 stratified train/test split. To deal with class imbalance, the Synthetic Minority Oversampling Technique (SMOTE) was applied only to the training data.

XGBoost achieved the highest accuracy of 93.17% and F1-Score of 93.46% after SMOTE, followed closely by Random Forest at 92.68% accuracy with a ROC-AUC of 97.38%. Logistic Regression and SVM achieved similar classification accuracy. The results substantiate the effectiveness of ensemble methods for early heart disease risk assessment and clinical decision-making.

Keywords: *heart disease prediction, early diagnosis, UCI Cleveland dataset, comparative analysis, machine learning, classification, SMOTE, XGBoost, Random Forest, clinical decision support.*

1. INTRODUCTION

The prevalence of heart disease has increased significantly in recent years, making it one of the leading causes of death worldwide. According to the World Heart Federation (WHF, 2025), obesity-related heart disease mortality has more than doubled over the past three decades, with women disproportionately affected. The World Health Organization's Global Hypertension Report (WHO, 2025) highlights that uncontrolled high blood pressure affects over 1 billion people and is a major cause of preventable death. Each year, approximately 17.9 million people worldwide die from heart disease.

Heart disease refers to various conditions affecting the heart and blood vessels including coronary artery disease, heart attack, arrhythmia, heart failure, irregular heartbeats, and congenital heart defects. Early detection and timely diagnosis of these conditions are crucial for reducing mortality and improving patient outcomes. Research has shown that early detection and diagnosis can reduce cardiovascular mortality by up to 30%, thereby improving survival rates and quality of life.

However, current methods of diagnosis often involve expensive and time-consuming procedures such as electrocardiograms and biochemical tests that require expertise not always available in resource-scarce settings. Inconsistency in diagnosis due to manual interpretation of patient records may also lead to delays in diagnosis and treatment. In developing nations, lifestyle factors such as inactive jobs, stress, unhealthy diets, and pollution lead to a rise in cases among younger adults, putting considerable strain on healthcare systems.

1.1 Problem Statement and Objectives

Although several studies have been conducted on heart disease prediction, many focus mainly on achieving high accuracy using a single model or without proper data balancing and metric comparison. The objective of this research is to perform a systematic comparison of multiple machine learning algorithms on the same dataset using proper preprocessing, class balancing, and medically relevant evaluation metrics such as recall and F1-score. This study also aims to:

- Determine the best-performing algorithm based on quantitative performance and clinical relevance, specifically minimizing false negatives.
- Assess the importance of significant clinical features in model predictions for interpretability in clinical settings.
- Evaluate the effect of SMOTE-based class balancing on each algorithm individually.

1.2 Scope of the Study

The main contributions of this study are: (1) a comparative study of the performance of five ML algorithms — Logistic Regression, Decision Tree, Random Forest, SVM, and XGBoost — under the same preprocessing environment; (2) evaluation of model performance with and without class

balancing using four clinically relevant performance metrics; and (3) analysis of feature importance of clinical parameters affecting heart disease prediction.

2. METHODOLOGY

2.1 Dataset

The dataset used in this study is the publicly available heart disease dataset obtained from the UCI Machine Learning Repository. It contains 1025 patient records with 13 clinical and demographic features. The target variable is binary, indicating the presence or absence of heart disease. The dataset contains medical information of patients related to heart health as described in Table I.

Table I — Key Attributes of Heart Disease Dataset

S. No.	Attribute	Description
1	Age	Age of the patient
2	Sex	Gender of the patient
3	cp	Chest pain type
4	trestbps	Resting blood pressure (mm Hg)
5	chol	Serum cholesterol level
6	fbs	Fasting blood sugar
7	restecg	Resting electrocardiographic results
8	thalach	Maximum heart rate achieved
9	exang	Exercise-induced angina
10	oldpeak	ST depression induced by exercise relative to rest
11	slope	Slope of the peak exercise ST segment
12	thal	Thalassemia condition

13	target	Heart disease status (binary)

2.2 Data Preprocessing

Data preprocessing is a critical step in ensuring the quality and reliability of the dataset. The following steps were performed:

- Missing values were replaced by mean imputation; duplicate records were identified and removed.
- Categorical attributes were transformed into numerical representations suitable for machine learning algorithms.
- All features were scaled using StandardScaler to normalize the feature range.
- The dataset was split into training (80%) and testing (20%) subsets using stratified sampling to preserve class proportions.

2.3 Class Balancing with SMOTE

The UCI Dataset exhibits a moderate class imbalance, with 54.5% of samples belonging to the positive class (heart disease present) and 45.5% to the negative class. The Synthetic Minority Oversampling Technique (SMOTE) was applied exclusively to the training set to address the imbalance. SMOTE generates synthetic samples for the minority class, thereby balancing the class distribution without discarding majority class samples.

2.4 Evaluation Metrics

The effectiveness of each model was measured using the following standard evaluation metrics:

- Accuracy: Fraction of total correct predictions.
- Precision: Fraction of true positives among all predicted positives.
- Recall: Fraction of true positives among all actual positives (most critical in clinical settings).
- F1-Score: Harmonic mean of Precision and Recall.
- ROC-AUC: Area under the Receiver Operating Characteristic curve measuring discriminative ability.

3. TOOLS AND TECHNOLOGIES

3.1 Programming Language and Environment

Python 3.x was used as the primary programming language for this study due to its extensive ecosystem of machine learning and data analysis libraries. Development was carried out in Jupyter Notebook, which provides an interactive environment suitable for iterative experimentation and result visualization.

3.2 Libraries and Frameworks

- NumPy and Pandas — for data manipulation, cleaning, and preprocessing of the UCI heart disease dataset.
- Scikit-learn (sklearn) — for implementing Logistic Regression, Decision Tree, Random Forest, and SVM classifiers; also used for StandardScaler, train-test split, and evaluation metrics (accuracy_score, classification_report, roc_auc_score).
- XGBoost — an optimized gradient boosting library used for implementing the XGBoost classifier.
- imbalanced-learn (imblearn) — for applying the SMOTE oversampling technique to the training data.
- Matplotlib and Seaborn — for generating visualizations including bar charts, confusion matrices, and ROC-AUC curves.

3.3 Dataset Source

The heart disease dataset was obtained from the UCI Machine Learning Repository (https://archive.ics.uci.edu/ml/datasets/heart+disease), which is one of the most widely used benchmark repositories for machine learning research. The specific version used contains 1025 records with 13 clinical attributes and a binary target variable.

3.4 Hardware Configuration

All experiments were conducted on a standard laptop system with an Intel Core i5 processor (or equivalent), 8 GB RAM, and Windows 10 operating system. No GPU acceleration was required, as the dataset size and model complexity were within the manageable range for CPU-based computation.

4. IMPLEMENTATION

4.1 Machine Learning Algorithms

Five supervised machine learning classifiers were implemented and evaluated. Each classifier was configured with specific parameters as described below.

Logistic Regression

Logistic Regression is a supervised machine learning algorithm used for binary classification. It estimates the probability of an event occurring using the relationship between dependent and independent variables with the help of the sigmoid function. Parameter used: max_iter=1000, allowing sufficient optimization iterations for convergence.

Decision Tree

A Decision Tree uses a tree-like structure where each internal node represents a decision based on features, branches represent results of those decisions, and leaves represent class labels. This model is non-parametric, requires minimal preprocessing, and offers high interpretability — ideal for clinical settings. Overfitting was addressed by adjusting the depth of the tree (random_state=42).

Random Forest

Random Forest is an ensemble of decision trees where predictions are combined through majority voting. Randomness in data selection (bagging) and feature selection at each node reduces variance and overfitting. It also calculates feature importance scores for clinical interpretability. Parameters: n_estimators=100, random_state=42.

Support Vector Machine (SVM)

SVM is a discriminative classifier that finds the optimal hyperplane maximizing the margin between classes in a high-dimensional feature space. The RBF kernel function was used, as it performs better on medical data with non-linear decision boundaries. Parameters: kernel=rbf, probability=True, random_state=42.

XGBoost

XGBoost is an optimized distributed gradient boosting library that builds an ensemble of decision trees sequentially, with each tree trained to correct the errors of the previous one. It incorporates L1 and L2 regularization to prevent overfitting and allows parallel tree construction for computational efficiency. Parameters: use_label_encoder=False, eval_metric=logloss.

4.2 Implementation Workflow

The implementation followed a structured pipeline:

1. Data Loading: Load the UCI heart disease dataset using Pandas.
2. Exploratory Data Analysis (EDA): Examine feature distributions, correlations, and class balance.
3. Data Preprocessing: Handle missing values, remove duplicates, encode categorical features, and apply StandardScaler.
4. Train-Test Split: Divide data into 80% training and 20% testing using stratified split.
5. SMOTE Application: Apply SMOTE to the training set only to address class imbalance.
6. Model Training: Train each of the five classifiers on both the original and SMOTE-balanced training sets.
7. Evaluation: Compute accuracy, precision, recall, F1-score, and ROC-AUC for each model on the test set.
8. Comparison: Generate comparison charts and confusion matrices to analyse performance differences.

5. MAJOR FINDINGS / OUTCOMES / OUTPUT / RESULTS

5.1 Performance Before SMOTE

Table II presents the complete performance metrics for all five algorithms before SMOTE application. XGBoost achieved the highest accuracy (93.66%) and F1-Score (93.95%), followed by SVM (92.68%) and Random Forest (91.71%). Logistic Regression showed the lowest accuracy at 80.98%.

Table II — Performance Before SMOTE

Model	Acc. (%)	Prec. (%)	Recall (%)	F1-Score (%)	ROC-AUC (%)
Logistic Regression	80.98	76.19	91.43	83.12	92.98
Decision Tree	87.32	86.24	89.52	87.85	93.26

Random Forest	91.71	89.29	95.24	92.17	97.23
SVM	92.68	91.67	94.29	92.96	97.71
XGBoost	93.66	91.82	96.19	93.95	97.95

5.2 Performance After SMOTE

Table III shows classifier performance after SMOTE-based class balancing. XGBoost achieved the highest accuracy (93.17%) and F1-Score (93.46%). Random Forest also demonstrated high accuracy (92.68%) and recall (95.24%). Notably, SVM accuracy dropped by 3.41 percentage points after SMOTE, making it the third-ranked model.

Table III — Performance After SMOTE

Model	Acc. (%)	Prec. (%)	Recall (%)	F1-Score (%)	ROC-AUC (%)
Logistic Regression	81.46	76.38	92.38	83.62	92.90
Decision Tree	87.32	86.24	89.52	87.85	93.26
Random Forest	92.68	90.91	95.24	93.02	97.38
SVM	89.27	86.73	93.33	89.91	96.77
XGBoost	93.17	91.74	95.24	93.46	97.44

5.3 Confusion Matrix Analysis

The confusion matrix (Table IV) reveals the true positive (TP), true negative (TN), false positive (FP), and false negative (FN) counts for each model on the test set. XGBoost and Random Forest both achieved the lowest false negative count (FN=5), indicating minimal missed heart disease cases — the most critical metric in clinical screening.

Table IV — Confusion Matrix

Model	TN	FP	FN	TP
Logistic Regression	70	30	8	97

Decision Tree	85	15	11	94
Random Forest	90	10	5	100
SVM	85	15	7	98
XGBoost	91	9	5	100

5.4 Model Ranking and Overfitting Analysis

Table V presents the final performance ranking of all five algorithms after SMOTE, ordered by accuracy, along with the overfitting gap (difference between training and test accuracy). A positive gap indicates mild overfitting; a negative gap indicates the model generalizes better on the test set than on training data.

Table V — Model Ranking After SMOTE

Model	Pre (%)	Recall (%)	F1 (%)	ROC-AUC (%)	Overfitting Gap
XGBoost	91.74	95.24	93.46	97.44	+0.0220
RF	90.91	95.24	93.02	97.38	+0.0325
SVM	86.73	93.33	89.91	96.77	+0.0079
DT	86.24	89.52	87.85	93.26	-0.0235
LR	76.38	92.38	83.62	92.90	-0.0238

5.5 Key Analysis and Discussion

The results of this study yield several important findings:

- XGBoost and Random Forest consistently outperformed all other models across all metrics. The performance difference between the two was marginal (0.49%), suggesting both models learn comparable representations of clinical data.
- SVM's accuracy declined by 3.41% after SMOTE. This occurs because SMOTE's synthetic samples may not accurately represent the true data distribution near the class boundary, causing SVM to identify sub-optimal support vectors.
- Decision Tree accuracy remained completely stable (87.32%) before and after SMOTE, indicating that tree-splitting criteria are robust to class rebalancing.
- Logistic Regression, despite its lowest accuracy (81.46%), achieved a recall of 92.38%, making it valuable as a high-sensitivity first-pass clinical screening tool.

- All five models achieved ROC-AUC values above 0.92, confirming strong discriminative ability across all classification thresholds.

6. CONCLUSION AND FUTURE SCOPE

6.1 Conclusion

This study performed a comprehensive comparative analysis of five machine learning algorithms — Logistic Regression, Decision Tree, Random Forest, SVM, and XGBoost — for heart disease prediction using the UCI Cleveland dataset with SMOTE-based class balancing. The key conclusions are as follows:

1. XGBoost achieved the highest accuracy of 93.17% and F1-score of 93.46% after SMOTE, making it the best-performing model overall.
2. Random Forest is the strongest alternative at 92.68% accuracy with a 97.38% ROC-AUC, virtually matching XGBoost's recall of 95.24%.
3. SMOTE improved most models but caused SVM to decline by 3.41%, confirming that SMOTE must be evaluated on a per-algorithm basis rather than applied blindly.
4. Logistic Regression's high recall (92.38%) makes it suitable as a first-pass clinical screening tool despite lower overall accuracy.
5. All five models achieved AUC above 0.92, confirming strong discriminative ability for heart disease classification.
6. XGBoost and Random Forest are recommended for clinical deployment based on their combined accuracy, recall, and AUC performance.

6.2 Future Scope

Several directions exist to extend and improve upon this work:

- Deep Learning Integration: Incorporating neural network architectures such as LSTM or CNN for more complex pattern recognition in clinical time-series data.
- Larger and Diverse Datasets: Validating results on larger, multicentric, and demographically diverse datasets to improve generalizability.
- Hyperparameter Optimization: Applying advanced techniques such as Bayesian optimization or genetic algorithms for systematic hyperparameter tuning.

REFERENCES

[1] S. Nikhar and A. M. Karandikar, "Prediction of heart disease using machine learning algorithms," International Journal of Advanced Engineering, Management and Science (IJAEMS), vol. 2, no. 6, pp. 617–623, Jun. 2016.

[2] B. Purushottama, K. Saxena and R. Sharma, "Efficient heart disease prediction system," Procedia Computer Science, vol. 85, pp. 962–969, 2016.

[3] C. M. Bhatt, P. Patel, T. Ghetia, and P. L. Mazzeo, "Effective heart disease prediction system using machine learning techniques," Algorithms, vol. 16, no. 2, p. 88, Feb. 2023.

[4] S. Mohan, C. Thirumalai, and G. Srivastava, "Effective heart disease prediction using hybrid machine learning techniques," IEEE Access, vol. 7, pp. 81542–81554, 2019.

[5] A. Lakshmanarao, A. Srisaila, and T. S. R. Kiran, "Heart disease prediction using feature selection and machine learning classification algorithms," International Journal of Electrical and Computer Engineering (IJECE), vol. 11, no. 1, pp. 46–53, 2021.

[6] R. Indrakumari, T. Poongodi, and B. Balamurugan, "Heart disease prediction using exploratory data analysis," Procedia Computer Science, vol. 173, pp. 130–139, 2020.

[7] M. Thenmozhi and P. C. Vimal, "Heart disease prediction using classification with feature selection," International Journal of Engineering Research and General Science, vol. 2, no. 3, ISSN 2091-2730, 2014.

[8] V. Chaurasia and S. Pal, "Early prediction of heart diseases using data mining techniques," Caribbean Journal of Science and Technology, vol. 1, pp. 208–217, 2013.

[9] L. Chen, "Heart disease prediction using machine learning techniques," Transactions on Materials, Biotechnology and Life Sciences, vol. 3, 2024.

[10] A. Garg, A. Sharma, and R. Khan, "Heart disease prediction using machine learning techniques," IOP Conference Series: Materials Science and Engineering, vol. 1022, 2021.

[11] J. Patel, T. Upadhyay, and S. Patel, "Predicting heart disease using classification mining technique," International Journal of Advance Research in Computer Science and Management Studies, vol. 7, pp. 129–137.